

KADI SARVA VISHWAVIDYALAYA**BE SEMESTER-VII Examination October - 2023****Subject Name: Distributed Systems****Subject Code: CT704A-N****Date: 02/11/2023****Time: 12:00pm to 03:00pm****Total Marks: 70****Instructions:**

1. Answer each section in separate answer sheet.
2. Use of scientific calculator is permitted.
3. All questions are Compulsory.
4. Indicate clearly, the option you attempt along with its respective question number.
5. Use the last page of main supplementary of rough work.

Section-I

- Q-1** (A) What is DS? Briefly discuss the various issues related to distributed system design. [5]
(B) Discuss Processor pool model and Workstation server model. [5]
(C) Explain distributed protocol FLIP with proper diagram. [5]

OR

- (C) Explain distributed protocol VMTP with VMTP packet format. [5]

- Q-2** (A) Explain different message buffering strategies. [5]
(B) Differentiate Centralize v/s Distributed computing. [5]

OR

- (A) Write and explain desired features of good process migration system. [5]
(B) Write short note on client-server binding. [5]

- Q-3** (A) What is RMI? Discuss the various components and the process of RMI execution. [5]
(B) What is the main difference between stateless and stateful servers? Which servers are used in distributed application? What is the role of persistent server? [5]

OR

- (A) Explain RPC communication protocols with necessary figure. [5]
(B) Discuss about ATM reference model with its functionality. [5]

Section-II

- Q-4** (A) What are various types of Hardware DSM? Explain them. [5]
(B) Explain different transparencies of distributed systems in detail. [5]
(C) How message can be ordered in IPC? Explain any two of them. [5]
- OR**
- (C) Explain failure handling mechanism of IPC? [5]
- Q-5** (A) What is deadlock? how it can be handled in distributed systems. [5]
(B) Explain how logical clocks are implemented in a distributed system. [5]
- OR**
- (A) What is clock skew? Briefly discuss about issues in clock synchronization. [5]
(B) Discuss various issues in implementing DSM systems. [5]
- Q-6** (A) Explain group communication mechanism. [5]
(B) Discuss and compare various election algorithms. [5]
- OR**
- (A) Explain different File models in Distributed File System. [5]
(B) Differentiate between Grid Computing and Cloud Computing. [5]